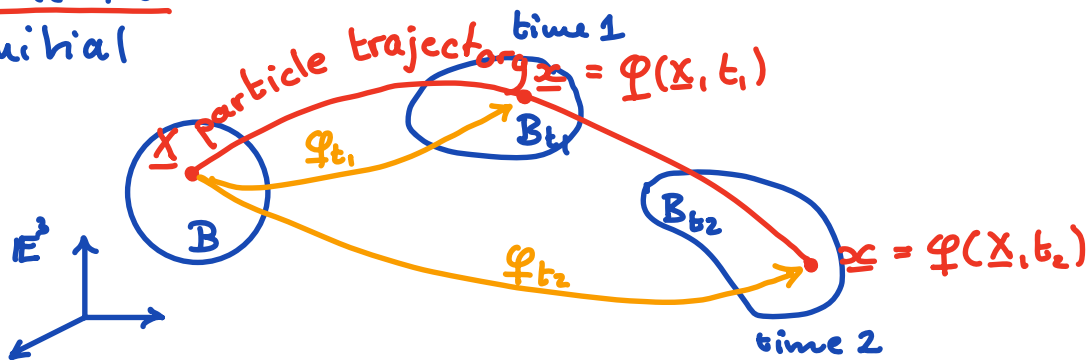


Motion

B : • undeformed
• reference
• material
• initial

B_t : • deformed
• spatial
• current



Continuous deformation of a body over time

is a motion: $\underline{x}(t) = \varphi_t(\underline{x}) = \varphi(\underline{x}, t)$

assume φ_t is admissible

Inverse motion: $\underline{x} = \underline{\psi}_t(\underline{x}) = \underline{\psi}(\underline{x}, t)$

Material and spatial fields

Temperature is naturally spatial field: $T(\underline{x}, t)$

Velocity is naturally material field: $\underline{v}(\underline{x}, t)$

φ_t & ψ_t : material $\longleftrightarrow$ spatial
 $\Rightarrow$ any field can be material or spatial

Material field is a field expressed in terms of
points $\underline{X} \in B$, e.g., $\Omega = \Omega(\underline{X}, t)$

Spatial field is a field expressed in terms of
points $\underline{x} \in B_t$, e.g., $\Gamma = \Gamma(\underline{x}, t)$

Spatial description of material field:

$$\Omega_s(\underline{x}, t) = \Omega(\psi(\underline{x}, t), t)$$

Material description of spatial field:

$$\Gamma_m(\underline{X}, t) = \Gamma(\varphi(\underline{X}, t), t)$$

Coordinate derivatives

?
Add definition in terms
of displacement
Holtz apfel p. 61 - 65

Material coordinates: $\nabla_{\underline{X}} = \text{Grad}, \text{Div}, \text{Curl}, \text{Lap}$

Spatial coordinates: $\nabla_{\underline{x}} = \text{grad}, \text{div}, \text{curl}, \text{lap}$

Velocity and acceleration fields

The velocity and acceleration of a material particle labeled $\underline{x}$ in $\mathcal{B}$ at time t due to motion $\varphi(\underline{X}, t)$

$$\begin{aligned}\underline{V}(\underline{X}, t) &= \frac{\partial}{\partial t} \varphi(\underline{X}, t) = \frac{\partial \underline{x}}{\partial t} \Big|_{\underline{X}} \\ \underline{A}(\underline{X}, t) &= \frac{\partial^2}{\partial t^2} \varphi(\underline{X}, t) = \frac{\partial^2 \underline{x}}{\partial t^2} \Big|_{\underline{X}}\end{aligned}$$

Spatial descriptions:

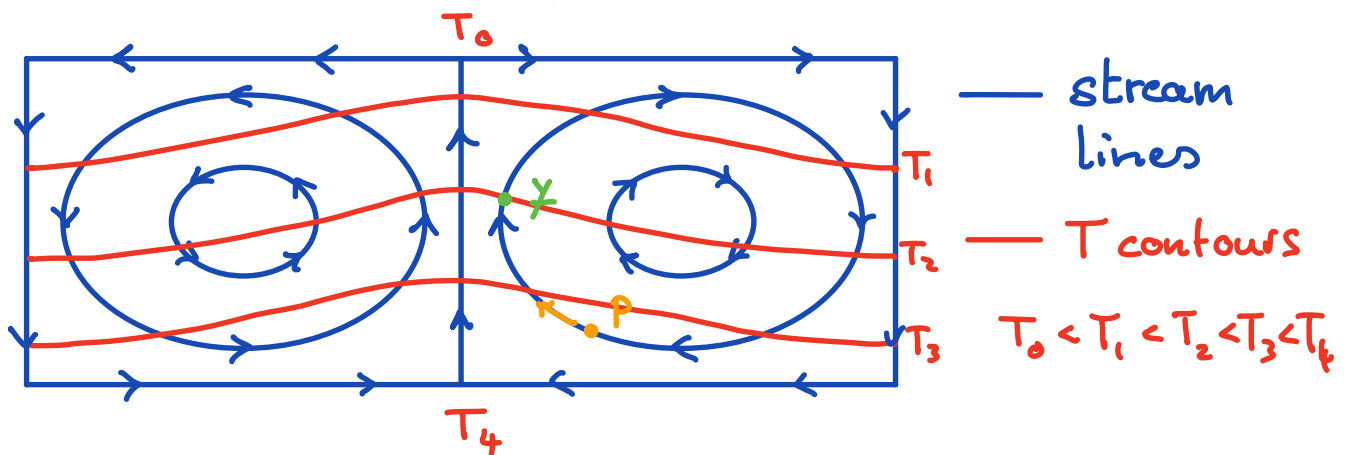
$$\underline{v}(\underline{x}, t) = \underline{V}_s(\underline{x}, t) = \frac{\partial}{\partial t} \varphi(\underline{\psi}(\underline{x}, t), t)$$

$$\underline{a}(\underline{x}, t) = \underline{A}_s(\underline{x}, t) = \frac{\partial^2}{\partial t^2} \varphi(\underline{\psi}(\underline{x}, t), t)$$

The spatial fields $\underline{v}$ and $\underline{a}$ correspond to the material particle whose current coordinates are $\underline{x}$ at time t .

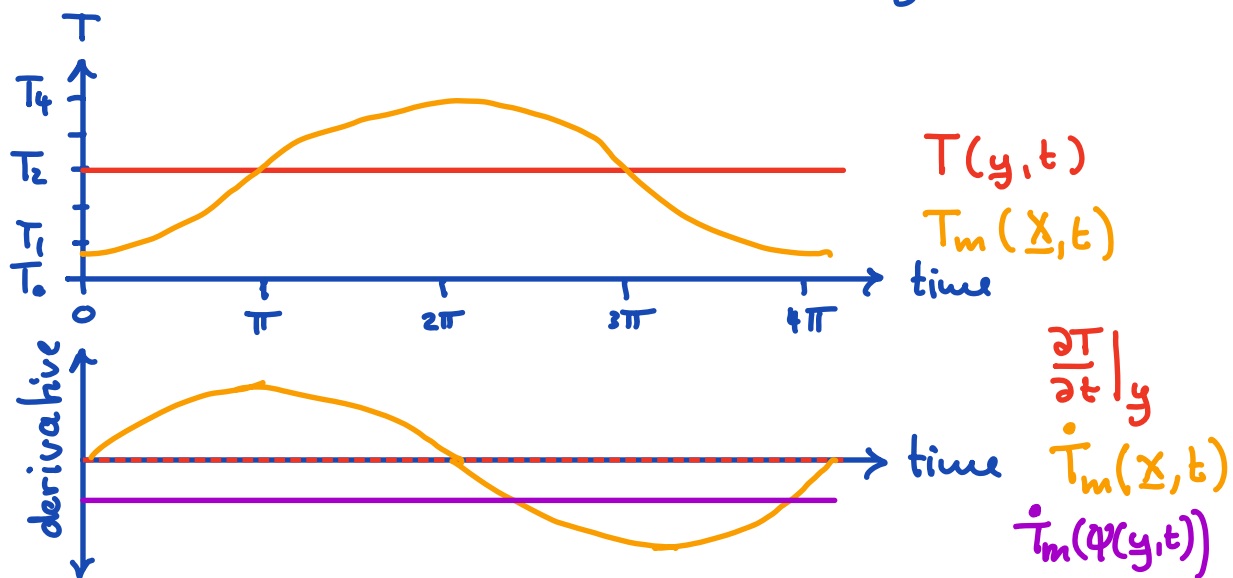
Note: Below we show that $\underline{a} \neq \frac{\partial \underline{v}}{\partial t}$!

Example: Steady convection



Steady state: $T(\underline{x}, t) = T(\underline{x})$ (spatial field)

Clearly, $T(y) = T_2$ and $\frac{\partial T}{\partial t} \Big|_y = 0$



Consider a particle P with initial location $\underline{x}$ and current position $\underline{x} = \varphi(\underline{x}, t)$, which passes

through y once every overturn.

Its temperature is $T(\varphi(\underline{x}, t)) = T_m(\underline{x}, t)$ is a material field and oscillates periodically. Its derivative the "material derivative" is $\dot{T}_m(\underline{x}, t) \neq 0$

Consider the particles passing through y with initial locations $\underline{y} = \underline{\psi}(\underline{y}, t)$. What is the change in temperature these particles experience as they pass through y ? $\dot{T}_m(\underline{\psi}(\underline{y}, t)) = [\dot{T}_m]_s$

Description of field

Time Derivative	Description of field	
	material	spatial
material/total	I $\dot{\Omega}(\underline{x}, t) = \frac{\partial \Omega}{\partial t} \Big _{\underline{x}}$	III $\dot{\Gamma}(\underline{x}, t) = \dot{\Omega}(\underline{\psi}(\underline{x}, t), t) = \dot{\Omega}_s(\underline{x}, t)$ $= \frac{\partial \Gamma}{\partial t} + \underline{v} \cdot \nabla \Gamma$
spatial/local	$\frac{\partial \Gamma}{\partial t}(\underline{\psi}(\underline{x}, t), t)$	II $\frac{\partial \Gamma}{\partial t} = \frac{\partial}{\partial t} \Gamma(\underline{x}, t) \Big _{\underline{x}}$

Different time derivatives

I) Material time derivative of material field Ω

Derivative of Ω with respect to t , holding $\underline{x}$ fixed.

$$\dot{\Omega}(\underline{x}, t) = \frac{D\Omega}{Dt}(\underline{x}, t) = \left. \frac{\partial \Omega}{\partial t} \right|_{\underline{x}}$$

Also called total, substantial or convective derivative.

$\dot{\Omega}$ represents the rate of change of Ω seen by an observer following the path line of a particle.

II) Spatial time derivative of a spatial field $\Gamma(\underline{x}, t)$

Derivative of Γ with respect to t , holding $\underline{x}$ fixed.

$$\left. \frac{\partial \Gamma}{\partial t}(\underline{x}, t) \right|_{\underline{x}} = \frac{\partial \Gamma}{\partial t}(\underline{x}, t)$$

Also referred to as the local time derivative

$\frac{\partial \Gamma}{\partial t}$ represents the rate of change in Γ as seen by an observer at $\underline{x}$.

III, Material time derivative of a spatial field

Derivative of scalar spatial field Γ with respect to time t , holding $\underline{X}$ fixed. The material coordinates $\underline{X}$ are fixed while the spatial coordinates change with time $\underline{x} = \underline{\varphi}(\underline{X}, t)$.

$$\dot{\Gamma}(\underline{x}, t) = \frac{D\Gamma}{Dt}(\underline{x}, t) = \frac{\partial}{\partial t} \underbrace{\Gamma(\underline{\varphi}(\underline{X}, t), t)}_{\Gamma_m(\underline{X}, t)} \Big|_{\underline{X} = \underline{\varphi}(\underline{x}, t)}$$

$\Rightarrow$ two time dependencies, one explicit the other implicit through the motion.

By the chain rule we have for fixed $\underline{X}$

$$\frac{\partial \Gamma}{\partial t}(\underline{\varphi}(\underline{X}, t), t) = \frac{\partial \Gamma}{\partial t}(\underline{x}, t) \Big|_{\underline{x} = \underline{\varphi}(\underline{X}, t)} + \frac{\partial \Gamma}{\partial x_i}(\underline{x}, t) \Big|_{\underline{x} = \underline{\varphi}(\underline{X}, t)} \frac{\partial}{\partial t} \varphi_i(\underline{X}, t)$$

recognizing spatial velocity: $v_i(\underline{x}, t) \Big|_{\underline{x} = \underline{\varphi}(\underline{X}, t)} = \frac{\partial}{\partial t} \varphi_i(\underline{X}, t)$

we have the total derivative

$$\frac{\partial \Gamma}{\partial t}(\underline{\varphi}(\underline{X}, t), t) = \left[\frac{\partial \Gamma}{\partial t}(\underline{x}, t) + \frac{\partial \Gamma}{\partial x_i}(\underline{x}, t) v_i(\underline{x}, t) \right]_{\underline{x} = \underline{\varphi}(\underline{X}, t)}$$

this is represented in material coords.

Expressing the result in terms of spatial coords

$$\dot{\Gamma}(\underline{x}, t) = \frac{\partial \Gamma}{\partial t}(\underline{x}, t) + \frac{\partial \Gamma}{\partial x_i}(\underline{x}, t) v_i(\underline{x}, t)$$

Let $\varphi(\underline{x}, t)$ be a motion with spatial velocity field $\underline{v}$ and consider spatial scalar $\phi = \phi(\underline{x}, t)$ and vector fields $\underline{\omega} = \underline{\omega}(\underline{x}, t)$. Then total time derivatives are given by

$$\dot{\phi} = \frac{\partial \phi}{\partial t} + \nabla^x \phi \cdot \underline{v} \quad \text{and} \quad \dot{\underline{\omega}} = \frac{\partial \underline{\omega}}{\partial t} + (\nabla^x \underline{\omega}) \underline{v}$$

The result for $\dot{\underline{\omega}}$ follows by applying the scalar result to ω_i .

This result is important because it allows the computation of $\dot{\phi}$ and $\dot{\underline{\omega}}$ without knowledge of φ , if the velocity is known!
 $\Rightarrow$ in fluid mechanics you never see φ

The spatial acceleration field is defined as

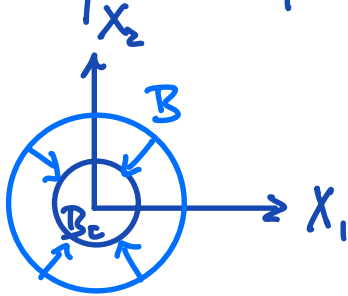
$$\underline{a} = \dot{\underline{v}} = \frac{\partial \underline{v}}{\partial t} + (\nabla^x \underline{v}) \underline{v}$$

Spatial acceleration is a non-linear function of spatial velocity $\Rightarrow$ basic non-linearity in fluid mechanics

Note: $\underline{a} \neq \frac{\partial \underline{v}}{\partial t}$! (only true for mat. fields $\underline{A} = \frac{\partial \underline{v}}{\partial t}$)

Finally, many texts, in particular in fluid mechanics, use the notation $\underline{v} \cdot \nabla^x \underline{\omega}$ instead of $(\nabla^x \underline{\omega}) \underline{v}$.

Example: Exponential expansion



$$\varphi(\underline{x}, t) = e^{-\lambda t} \underline{x}$$

$$\psi(\underline{x}, t) = e^{\lambda t} \underline{x}$$

Material fields:

$$\underline{V}(\underline{x}, t) = \frac{\partial}{\partial t} \varphi(\underline{x}, t) = -\lambda e^{-\lambda t} \underline{x}$$

$$\underline{A}(\underline{x}, t) = \frac{\partial}{\partial t} \underline{V}(\underline{x}, t) = \lambda^2 e^{-\lambda t} \underline{x}$$

Spatial fields:

$$\underline{v}(\underline{x}, t) = \underline{V}_s(\underline{x}, t) = \underline{V}(\psi(\underline{x}, t), t)$$

$$= -\lambda e^{-\lambda t} (e^{\lambda t} \underline{x}) = -\lambda \underline{x}$$

$$\underline{a}(\underline{x}, t) = \underline{A}(\psi(\underline{x}, t), t) = \lambda^2 e^{-\lambda t} (e^{\lambda t} \underline{x}) = \lambda^2 \underline{x}$$

Temperature field: $T_m(\underline{x}, t) = \alpha t \|\underline{x}\|$

Material time derivative:

$$\dot{T}_m = \frac{\partial}{\partial t} T_m = \alpha \|\underline{x}\|$$

(calculated directly from mat. field)

Spatial Temperature field:

$$T(\underline{x}, t) = T_m(\psi(\underline{x}, t), t) = \alpha t \|e^{\lambda t} \underline{x}\| = \alpha t \|\underline{x}\| e^{\lambda t}$$

Suppose we only know spatial fields

$T(\underline{x}, t)$ and $\underline{v}(\underline{x}, t)$. What is the material derivative $\dot{T}(\underline{x}, t)$?

Use $\dot{T} = \frac{\partial T}{\partial t} + \nabla_{\underline{x}} T \cdot \underline{v}$

$$\frac{\partial T}{\partial t} = \alpha \|\underline{x}\| e^{\lambda t} + \alpha \lambda t \|\underline{x}\| e^{\lambda t}$$

$$\nabla_{\underline{x}} T = \alpha t e^{\lambda t} \nabla_{\underline{x}} (\underline{x} \cdot \underline{x})^{\frac{1}{2}} = \alpha t e^{\lambda t} \frac{1}{2} (\underline{x} \cdot \underline{x})^{-\frac{1}{2}} \nabla_{\underline{x}} (\underline{x} \cdot \underline{x})$$

in components: $(x_i x_i)_{,j} = (x_i^2)_{,j} = 2 x_i x_{i,j} = 2 x_i \delta_{ij} = 2 x_j$

$$\Rightarrow \nabla_{\underline{x}} (\underline{x} \cdot \underline{x}) = 2 \underline{x}$$

$$\nabla_{\underline{x}} T = \alpha t e^{\lambda t} \frac{\underline{x}}{\|\underline{x}\|}$$

putting it all together:

$$\dot{T} = \underbrace{\alpha \|\underline{x}\| e^{\lambda t} + \alpha \lambda t \|\underline{x}\| e^{\lambda t}}_{\frac{\partial T}{\partial t}} + \underbrace{\alpha t e^{\lambda t} \frac{\underline{x}}{\|\underline{x}\|}}_{\nabla_{\underline{x}} T} \cdot \underbrace{(-\lambda \underline{x})}_{\underline{v}}$$

$$= \alpha \|\underline{x}\| e^{\lambda t} + \alpha \lambda t \|\underline{x}\| e^{\lambda t} - \alpha \lambda t \underbrace{\frac{\underline{x} \cdot \underline{x}}{\|\underline{x}\|}}_{\|\underline{x}\|} e^{\lambda t}$$

$$\dot{T}(\underline{x}, t) = \alpha \|\underline{x}\| e^{\lambda t} \quad \text{spatial description of mat. der.}$$

$$\dot{T}_m(\underline{X}) = \alpha \|\underline{X}\| \quad \rightarrow \quad \dot{T}_m(\underline{\psi}(\underline{x}, t)) = \alpha \|\underline{x}\| e^{\lambda t} = \dot{T}(\underline{x}, t)$$

$$\quad \quad \quad \uparrow$$

$$\underline{X} = \underline{\psi}(\underline{x}, t) = e^{\lambda t} \underline{x}$$